Mahidol University
Faculty of Information
and Communication Technology

System Startup, Linux File System, and Basic Command Line Operations

ITCS 254 | Lecture 3

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Agenda



Boot Process



Kernel, init, and Services



Linux Filesystem Basics



Getting Started with Basic Command Line Operations

Learning Objectives



Describe the boot process.



Identify Linux filesystems.

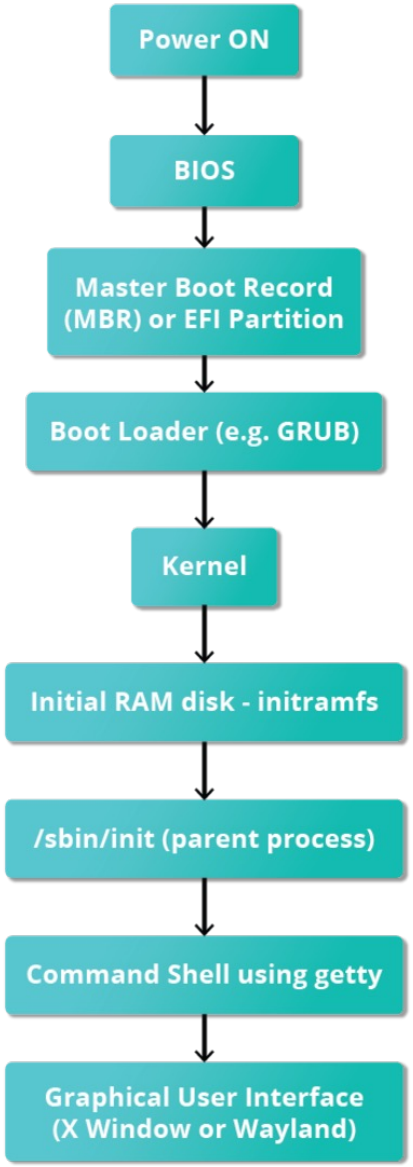


Identify the differences between partitions and filesystems.



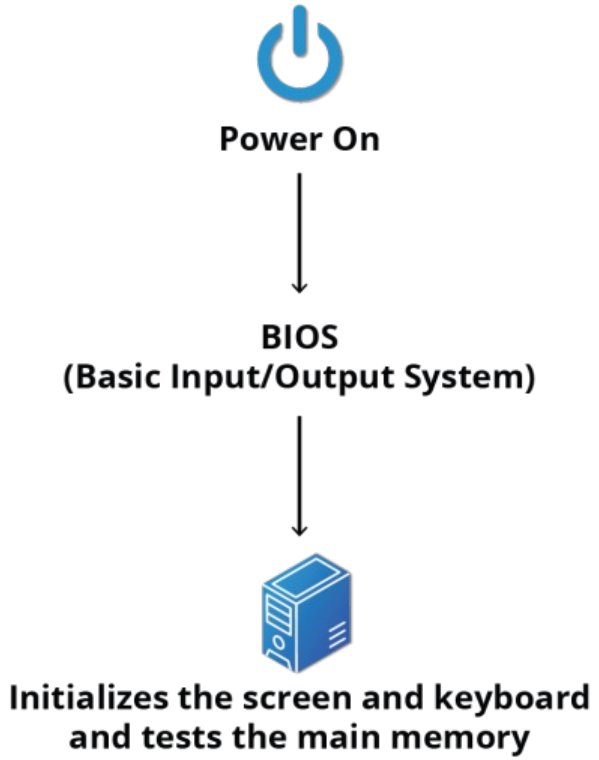
Use the command line to perform operations in Linux.

Boot Process Overview



BIOS

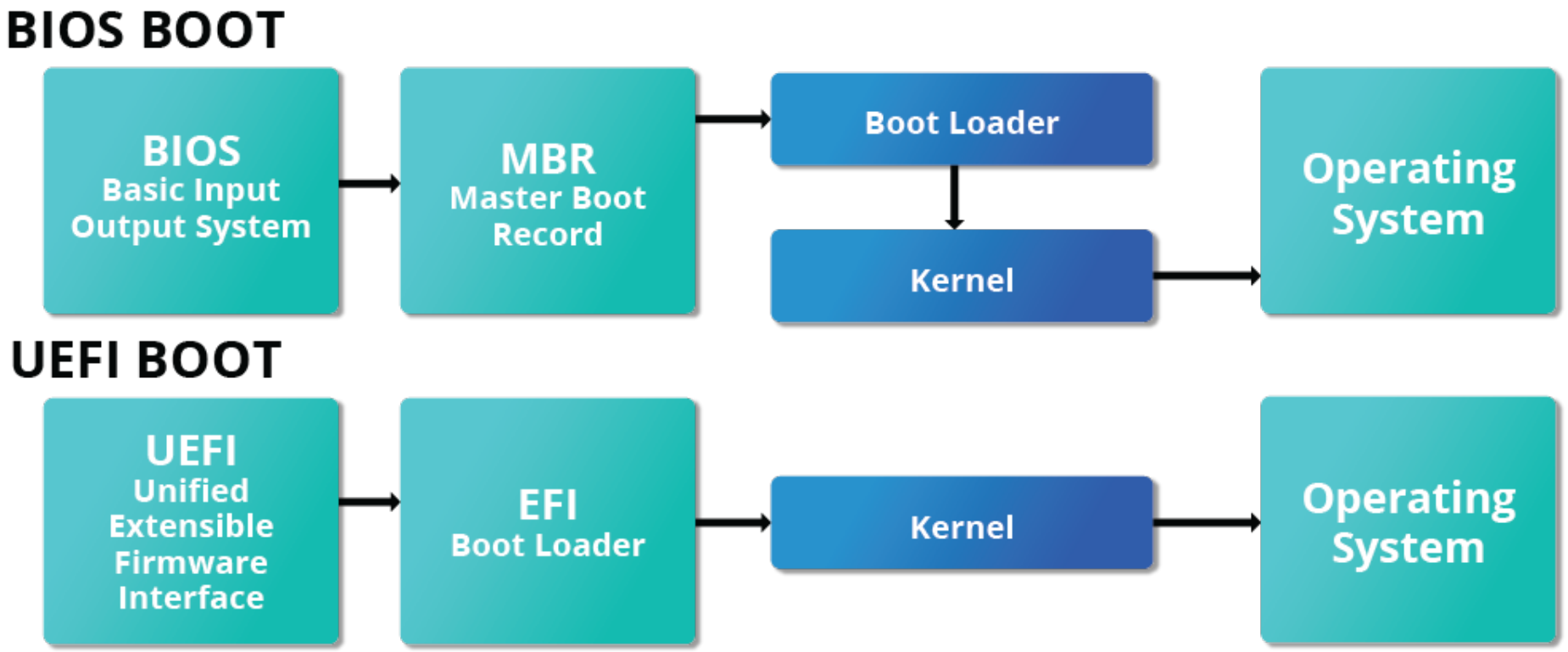
- When the computer is powered on, the Basic Input/Output System (BIOS) initializes the hardware, including the screen and keyboard, and tests the main memory.
  - This process is also called POST (Power On Self Test).
- The BIOS software is stored on a ROM chip on the motherboard.
- After this, the remainder of the boot process is controlled by the OS.
- Once the POST is completed, system control passes from the BIOS to the boot loader.



Master Boot Record (MBR), ERI Partition, and Boot Loader

- The boot loader is usually stored on one of the system's storage devices, such as a hard disk or SSD drive, either in the boot sector (for traditional BIOS/MBR systems) or the EFI partition (for more recent (Unified) Extensible Firmware Interface or EFI/UEFI systems).
- A number of boot loaders exist for Linux; the most common ones are GRUB (for GRand Unified Boot loader), ISOLINUX (for booting from removable media), and DAS U-Boot (for booting on embedded devices/appliances).
- Most Linux boot loaders can present a user interface for choosing alternative options for booting Linux and even other OSs that might be installed.
- When booting Linux, the boot loader is responsible for loading the kernel image and the initial RAM disk or filesystem (which contains some critical files and device drivers needed to start the system) into memory.

Master Boot Record (MBR), ERI Partition, and Boot Loader



Boot Loader in Action: The First Stage

- For systems using the BIOS/MBR method
  - the boot loader resides at the first sector of the hard disk, also known as the MBR. The size of the MBR is just 512 bytes.
  - The boot loader examines the partition table and finds a bootable partition.
  - Once it finds a bootable partition, it then searches for the second stage boot loader, such as GRUB, and loads it into RAM.
- For systems using the EFI/UEFI method
  - UEFI firmware reads its Boot Manager data to determine which UEFI application is to be launched and from where (i.e., from which disk and partition the EFI partition can be found).
  - The firmware then launches the UEFI application, such as GRUB, as defined in the boot entry in the firmware's boot manager.
  - This procedure is more complicated but more versatile than the older MBR methods.

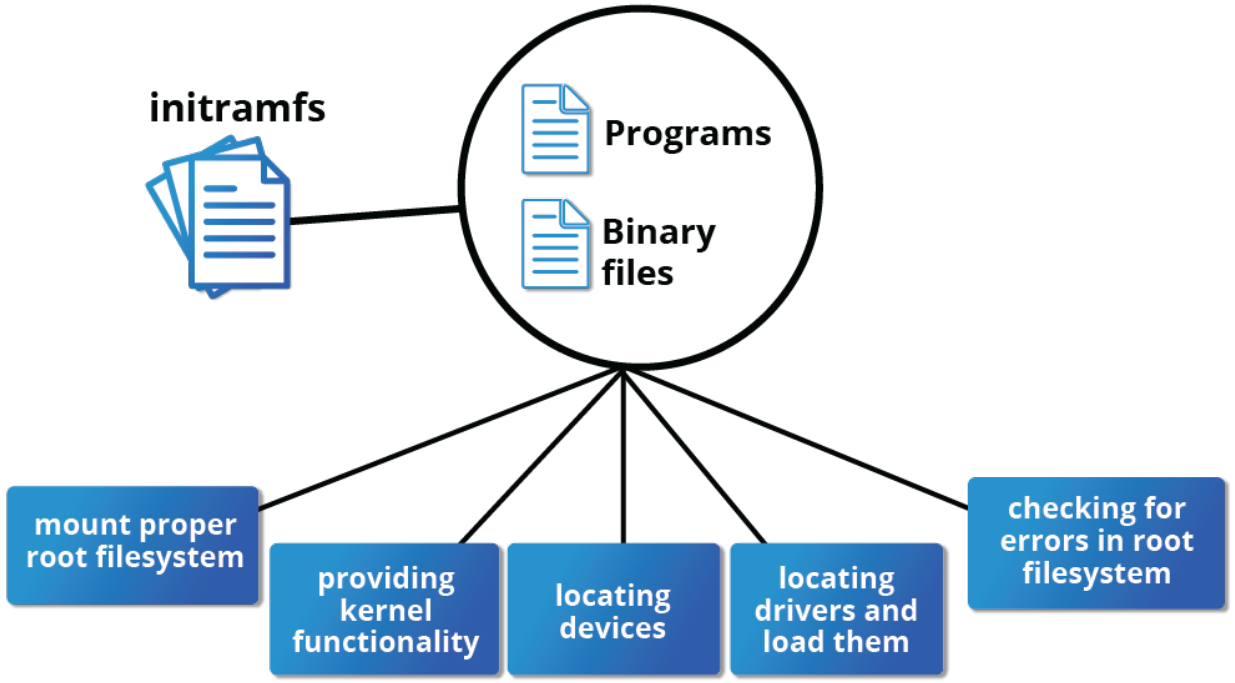
Boot Loader in Action: The Second Stage

- Boot loader resides under `/boot`.
- A splash screen is displayed, which allows us to choose which OS and/or kernel to boot.
- After the OS and kernel are selected, the boot loader loads the kernel of the OS into RAM and passes control to it.
- Kernels are almost always compressed, so the first job they have is to uncompress themself.
- It will check and analyze the system hardware and initialize any hardware device drivers built into the kernel.



Source: https://askubuntu.com/questions/1446821/grub-bootloader-doesnt-show-windows-7

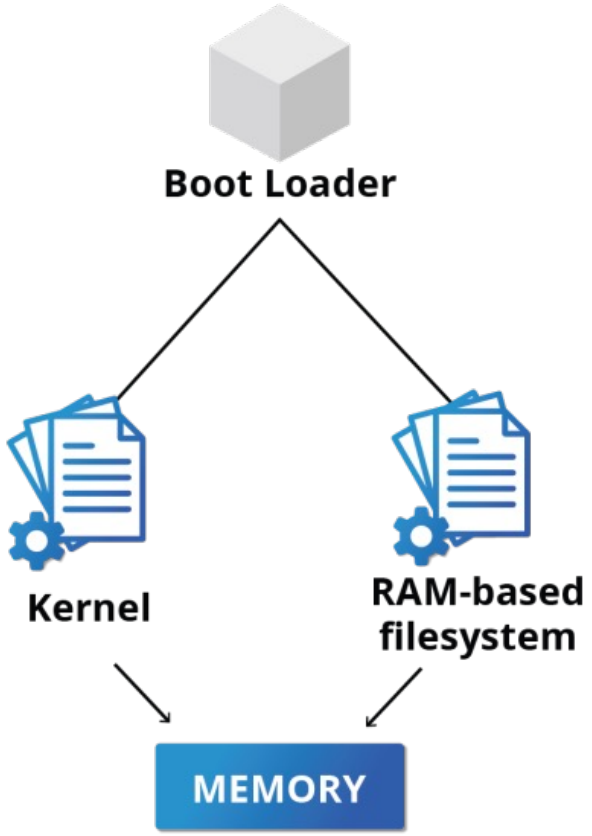
The Initial RAM Disk



- The mount program instructs the OS that a filesystem is ready for use and associates it with a particular point in the overall hierarchy of the filesystem (the mount point).
- If this is successful, the `initramfs` is cleared from RAM, and the `init` program on the root filesystem (`/sbin/init`) is executed.
- `init` handles the mounting and pivoting over to the final real root filesystem.
- If special hardware drivers are needed before the mass storage can be accessed, they must be in the `initramfs` image.

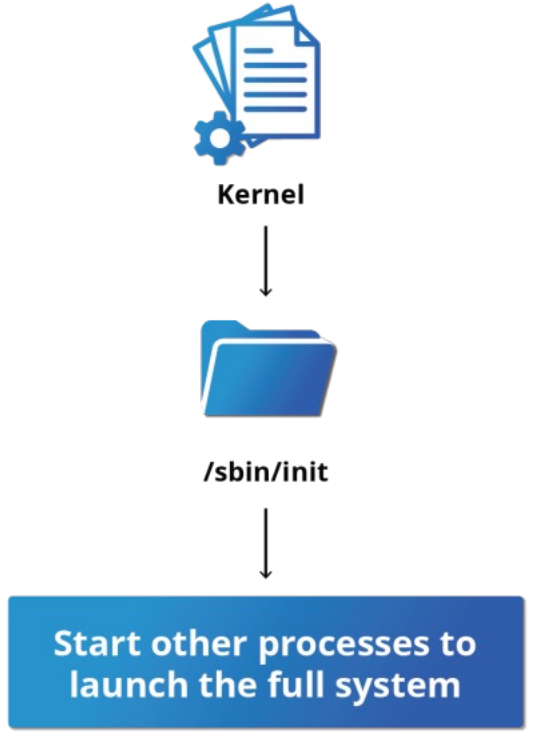
The Linux Kernel

- The boot loader loads both the kernel and an initial RAM-based file system (`initramfs`) into memory, so it can be used directly by the kernel.
- When the kernel is loaded in RAM, it immediately initializes and configures the computer's memory and configures all the hardware attached to the system.
  - This includes all processors, I/O subsystems, storage devices, etc.
- The kernel also loads some necessary user space applications.



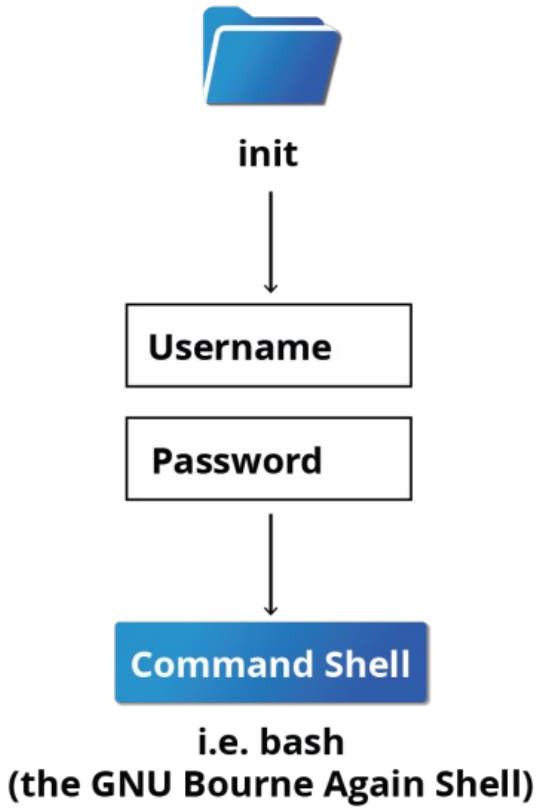
`/sbin/init` and Services

- Once the kernel has set up all its hardware and mounted the root filesystem, the kernel runs `/sbin/init`.
- This then becomes the initial process, which then starts other processes to get the system running.
- Most other processes on the system trace their origin ultimately to `init`; exceptions include the so-called kernel processes.
- Besides starting the system, `init` is responsible for keeping the system running and for shutting it down cleanly.
- One of its responsibilities is to act when necessary as a manager for all non-kernel processes; it cleans up after them upon completion, and restarts user login services as needed when users log in and out, and does the same for other background system services.



Text-Mode Login

- Near the end of the boot process, `init` starts a number of text-mode login prompts.
  - If you are running a system with a graphical login interface, you will not see these at first.
- Usually, the default command shell is `bash`, but there are a number of other advanced command shells available.



systemd

- Systems with `systemd` start up faster than those with earlier `init` methods.
- This is largely because it replaces a serialized set of steps with aggressive parallelization techniques, which permits multiple services to be initiated simultaneously.
- Complicated startup shell scripts are replaced with simpler configuration files, which enumerate what has to be done before a service is started, how to execute service startup, and what conditions the service should indicate have been accomplished when startup is finished.
- One thing to note is that `/sbin/init` now just points to `/lib/systemd/systemd`; i.e., `systemd` takes over the `init` process.
- One `systemd` command (`systemctl`) is used for most basic tasks.

```
$ sudo systemctl start|stop|restart httpd.service
```

Linux Filesystems

- Libraries separate books and other media into multiple sections; this organization will depend on the subject matter, audience, media type, and frequency of retrieval.
- The same concept applies to a filesystem, which is the embodiment of a method of storing and organizing arbitrary collections of data in a human-usable form.
- Different types of filesystems supported by Linux
  - Conventional disk filesystems: ext3, ext4, XFS, Btrfs, JFS, NTFS, vfat, exfat, etc.
  - Flash storage filesystems: ubifs, jffs2, yaffs, etc.
  - Database filesystems
  - Special purpose filesystems: procfs, sysfs, tmpfs, squashfs, debugfs, fuse, etc.

Partitions and Filesystems

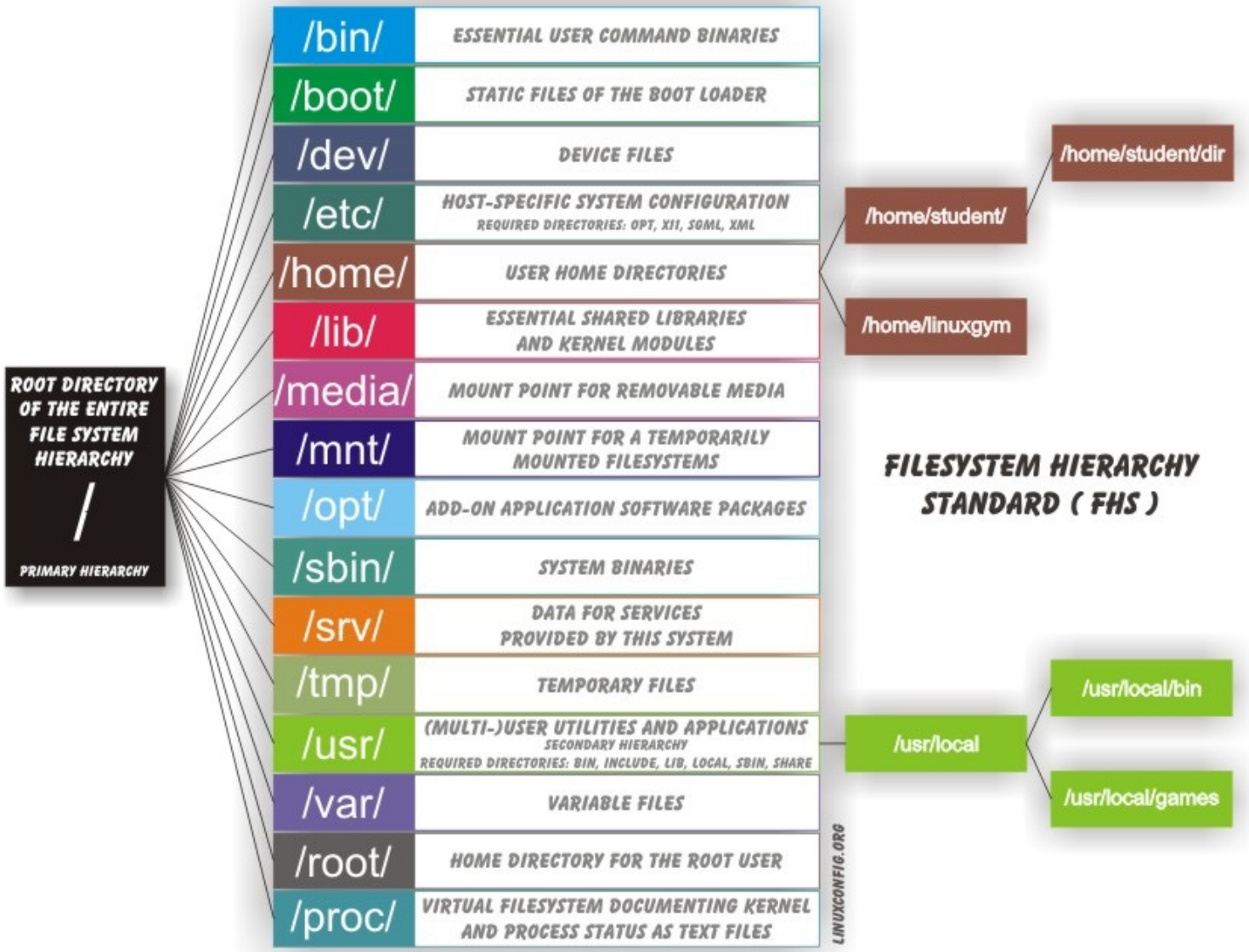
- A partition is a dedicated subsection of physical storage media.
- A filesystem is just a method of storing and accessing files.
- One can think of a partition as a container in which a filesystem resides. However, in some circumstances, a filesystem can span more than one partition if one uses symbolic links.

	Windows	Linux
Partition	Disk1	/dev/sda1
Filesystem type	NTFS/VFAT	EXT3/EXT4/XFS/BTRFS…
Mounting parameters	DriveLetter	MountPoint
Base folder (where OS is stored)	C:\	/

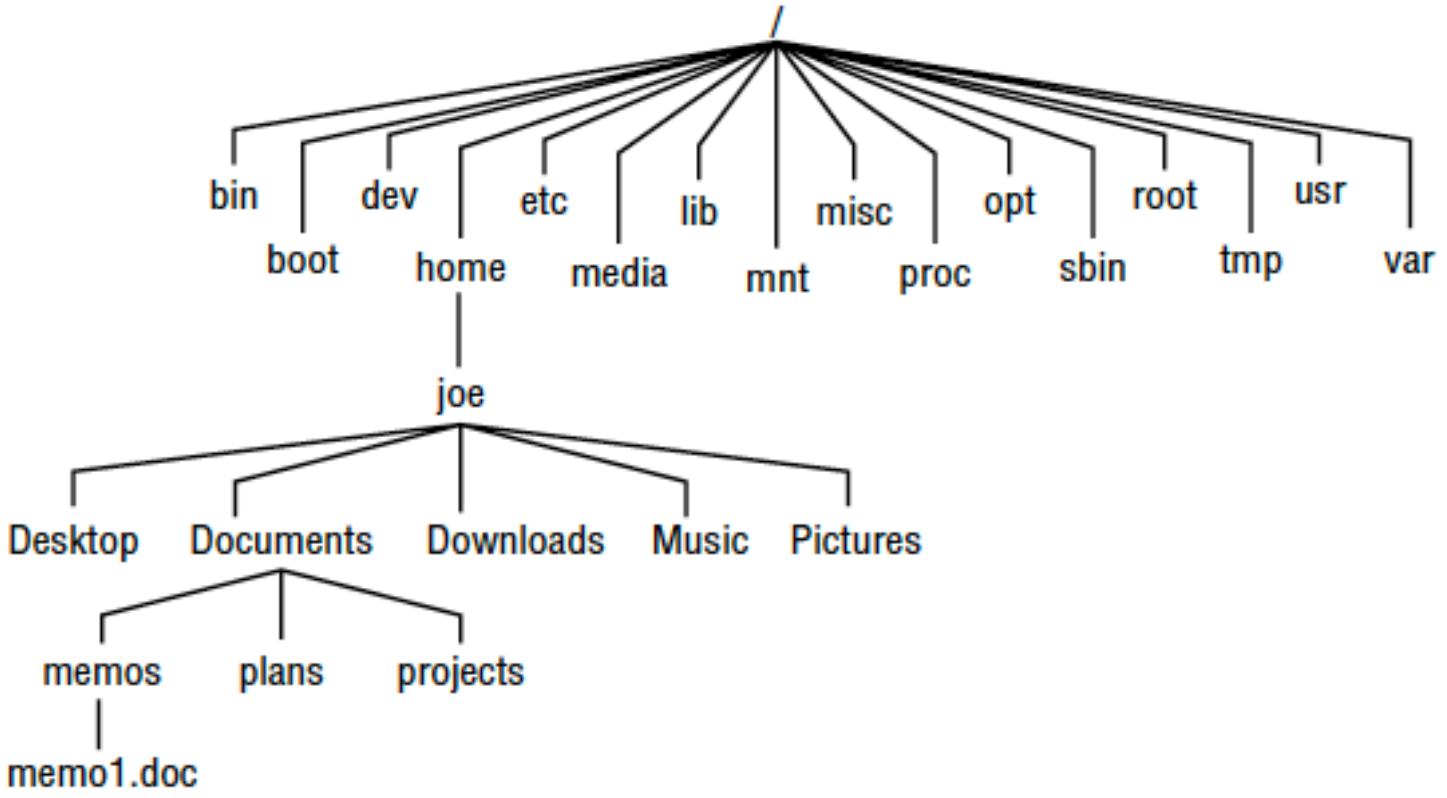
The Filesystem Hierarchy Standard

- Linux systems store their important files according to a standard layout called the Filesystem Hierarchy Standard (FHS), which has long been maintained by the Linux Foundation.
- https://refspecs.linuxfoundation.org/FHS_3.0/fhs-3.0.pdf
- Having a standard is designed to ensure that users, administrators, and developers can move between distributions without having to re-learn how the system is organized.
- Multiple drives and/or partitions are mounted as directories in the single filesystem.
- Removable media such as USB drives and CDs, and DVDs will show up as mounted at `/run/media/`*`username`*`/`*`disklabel`* for recent Linux systems or under `/media` for older distributions.
- All Linux filesystem names are case-sensitive.

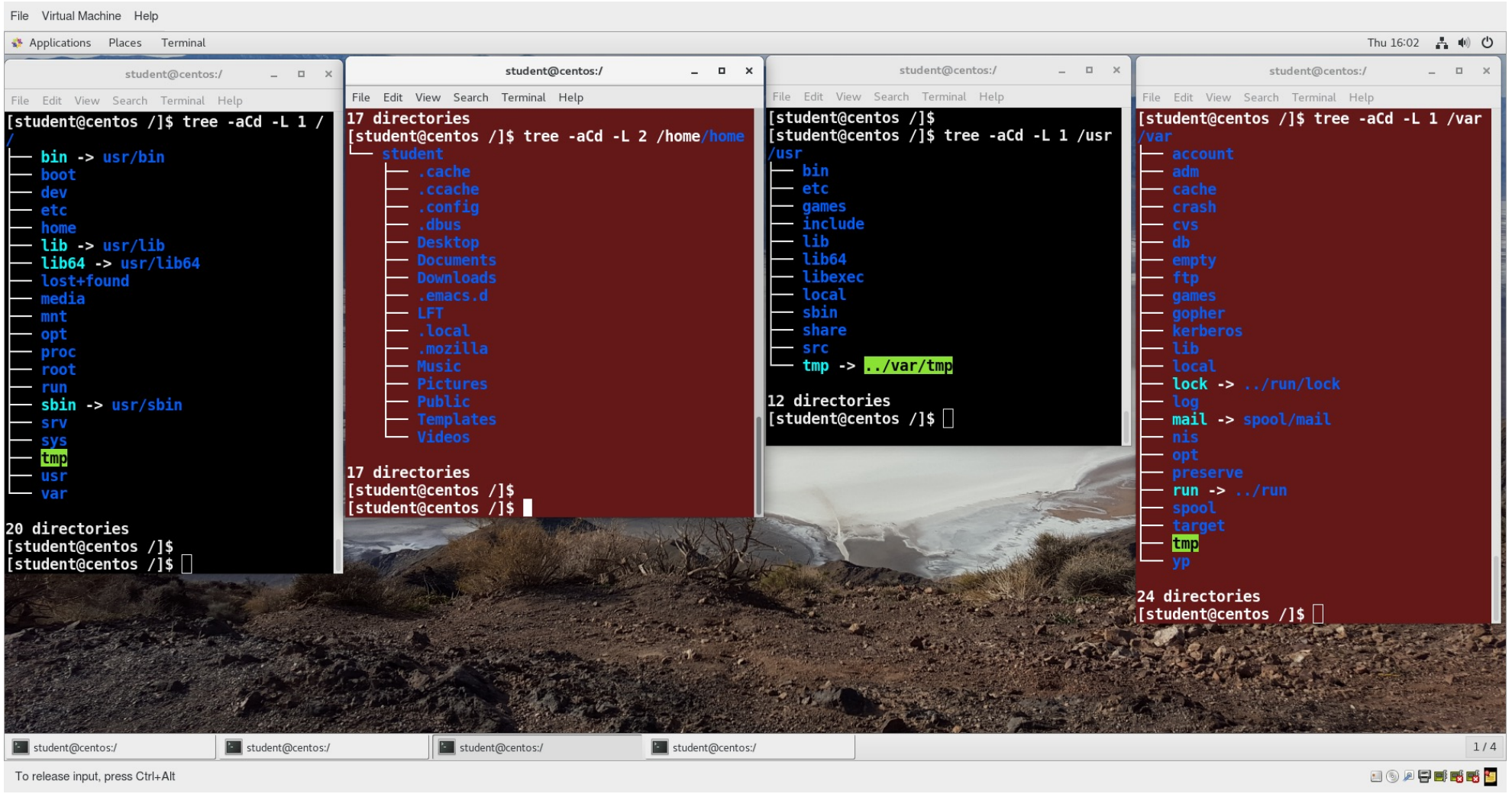
The Filesystem Hierarchy Standard



The Linux Filesystem is Organized as a Hierarchy



Filesystem Hierarchy Standard



Introduction to the Command Line



- No GUI overhead is incurred.
- Virtually any and every task can be accomplished while sitting at the command line.
- You can implement scripts for often-used (or easy-to-forget) tasks and series of procedures.
- You can sign into remote machines anywhere on the Internet.
- You can initiate graphical applications directly from the command line instead of hunting through menus.
- While graphical tools may vary among Linux distributions, the command line interface does not.

Using a Text Terminal on the Graphical Desktop

- By default, on GNOME desktop environments, the `gnome-terminal` application is used to emulate a text-mode terminal in a window.
- Other available terminal programs include: xterm, konsole (default on KDE), terminator.
- To open a terminal on any system using a recent GNOME desktop click on Applications > System Tools > Terminal or Applications > Utilities > Terminal.
- On most GNOME-based distributions, you can always open a terminal by right-clicking anywhere on the desktop background and selecting Open in Terminal.
  - If this does not work, you may need to install and activate the appropriate package, usually called gnome-shell-extension-apps-menu.
- You can also hit Alt-F2 and type `gnome-terminal`.

The Command Line

- Most input lines entered at the shell prompt have three basic elements

```
$ command [-options] [--long_option] [arguments]
```

- The command is the name of the program or script you are executing.
- It may be followed by one or more options (or switches) that modify what the command may do.
  - Options usually start with one or two dashes, such as, `-p` or `--print`, in order to differentiate them from arguments, which represent what the command operates on.
- Plenty of commands have no options, no arguments, or neither.
- Other elements (such as setting environment variables) can also appear on the command line when launching a task.

Using The Shell Prompt

- The default prompt for a regular user: $
- The default prompt for the root user: #
- In most Linux systems, the $ and # prompts are preceded by your username, system name, and current directory name.

```
[ittipon@dbserver ~] $
```

Understanding Command Syntax

```
$ ls

$ ls /usr

$ ls -l

$ ls -l -a

$ ls -la

$ ls --hide=Desktop

$ ls -I Desktop

$ man ls
```

Disable/Enable GUI in Ubuntu

- Disable GUI

```
$ sudo systemctl set-default multi-user
```

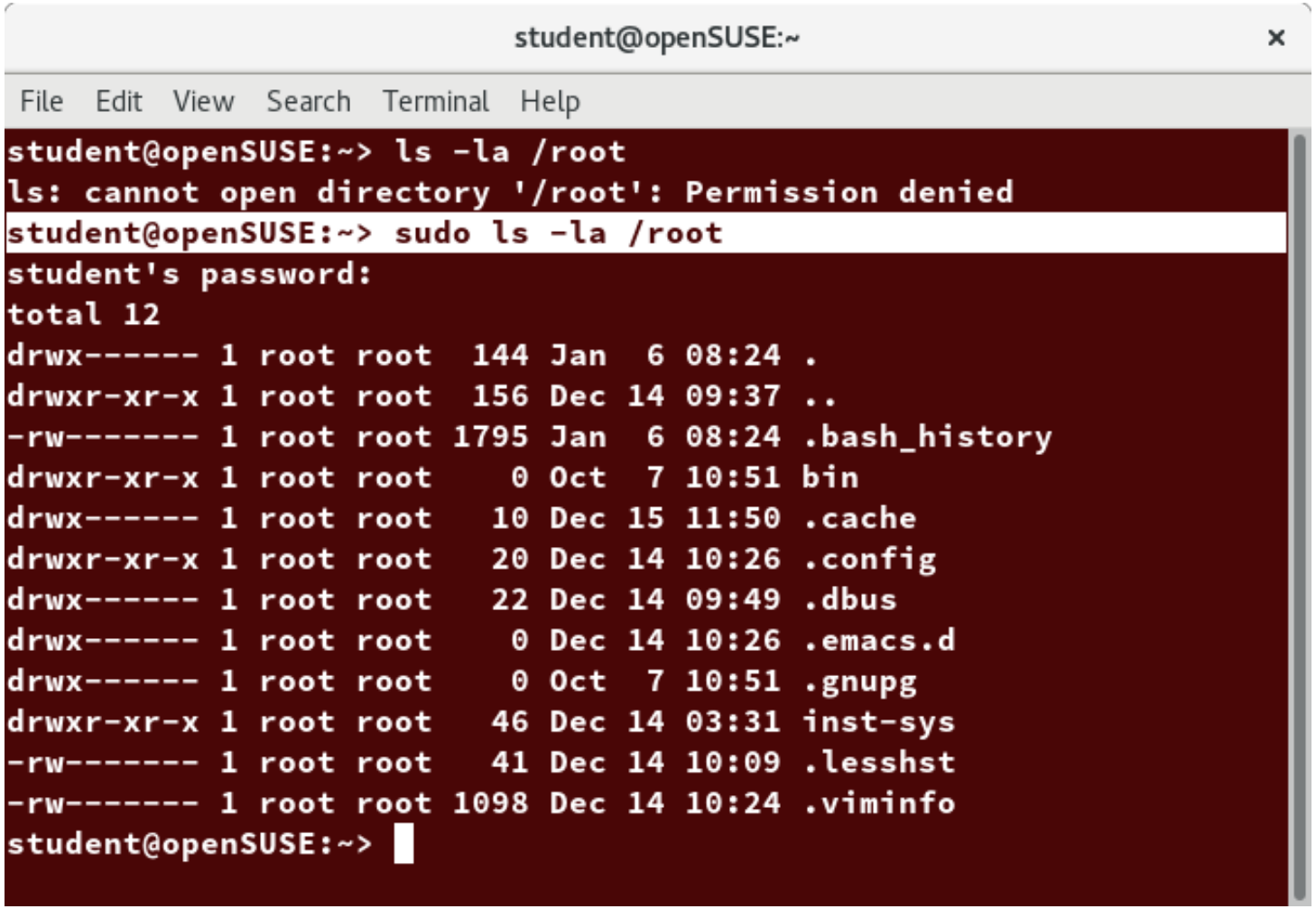
```
$ reboot (or gnome-session-quit)
```

- Enable GUI

```
$ sudo systemctl set-default graphical
```

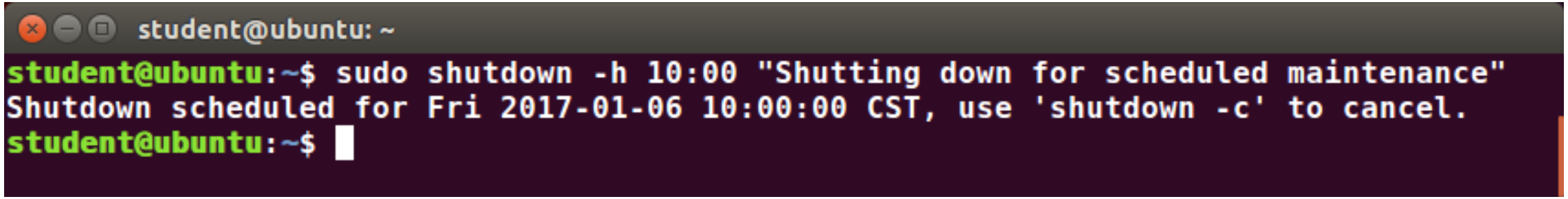
```
$ sudo systemctl start gdm3
```

sudo


student@openSUSE:~
File Edit View Search Terminal Help
student@openSUSE:~> ls -la /root
ls: cannot open directory '/root': Permission denied
student@openSUSE:~> sudo ls -la /root
student's password:
total 12
drwx------ 1 root root 144 Jan 6 08:24 .
drwxr-xr-x 1 root root 156 Dec 14 09:37 ..
-rw------- 1 root root 1795 Jan 6 08:24 .bash_history
drwxr-xr-x 1 root root   0 Oct 7 10:51 bin
drwx------ 1 root root   10 Dec 15 11:50 .cache
drwxr-xr-x 1 root root   20 Dec 14 10:26 .config
drwx------ 1 root root   22 Dec 14 09:49 .dbus
drwx------ 1 root root   0 Dec 14 10:26 .emacs.d
drwx------ 1 root root   0 Oct 7 10:51 .gnupg
drwxr-xr-x 1 root root   46 Dec 14 03:31 inst-sys
-rw------- 1 root root   41 Dec 14 10:09 .lesshst
-rw------- 1 root root 1098 Dec 14 10:24 .viminfo
student@openSUSE:~>

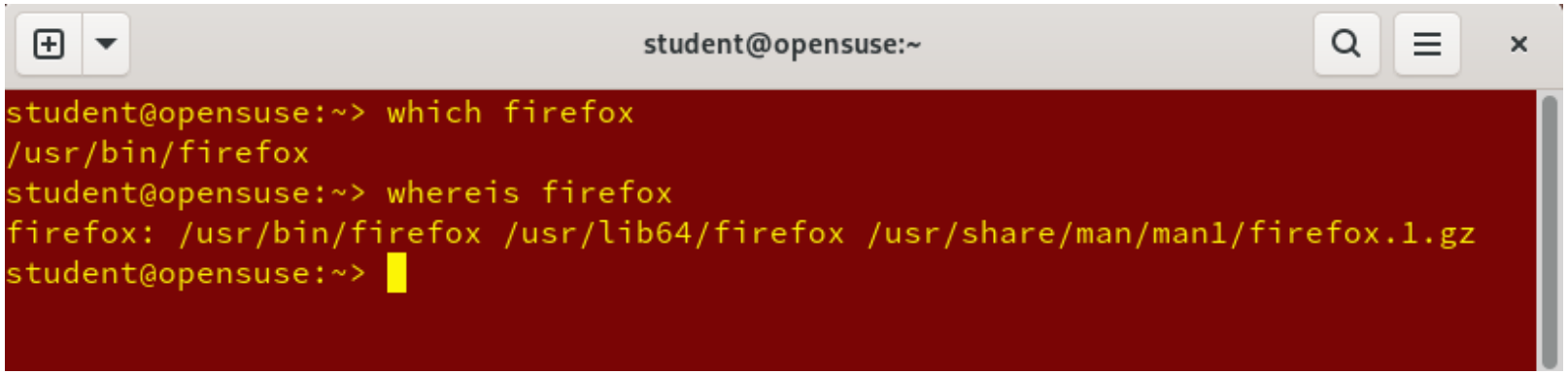
Rebooting and Shutting Down

- `$ shutdown -r now`
- `$ shutdown -r`
- `$ shutdown -h now`
- `$ shutdown -h`



Locating Applications

- In general, executable programs and scripts should live in the `/bin`, `/usr/bin`, `/sbin`, `/usr/sbin` directories, or somewhere under `/opt`.
- They can also appear in `/usr/local/bin` and `/usr/local/sbin`, or in a directory in a user's account space, such as `/home/student/bin`.
- `$ which diff`
- `$ whereis diff`



Accessing Directories

- `cd` Change the current directory to another specific directory.
- `cd ~` Change to your home directory; shortcut name is ~ (tilde).
- `cd $HOME` Change to your home directory
- `cd ..` Change to parent directory (`..`)
- `cd -` Change to previous working directory; - (minus)
- `pwd` Displays the present working directory.

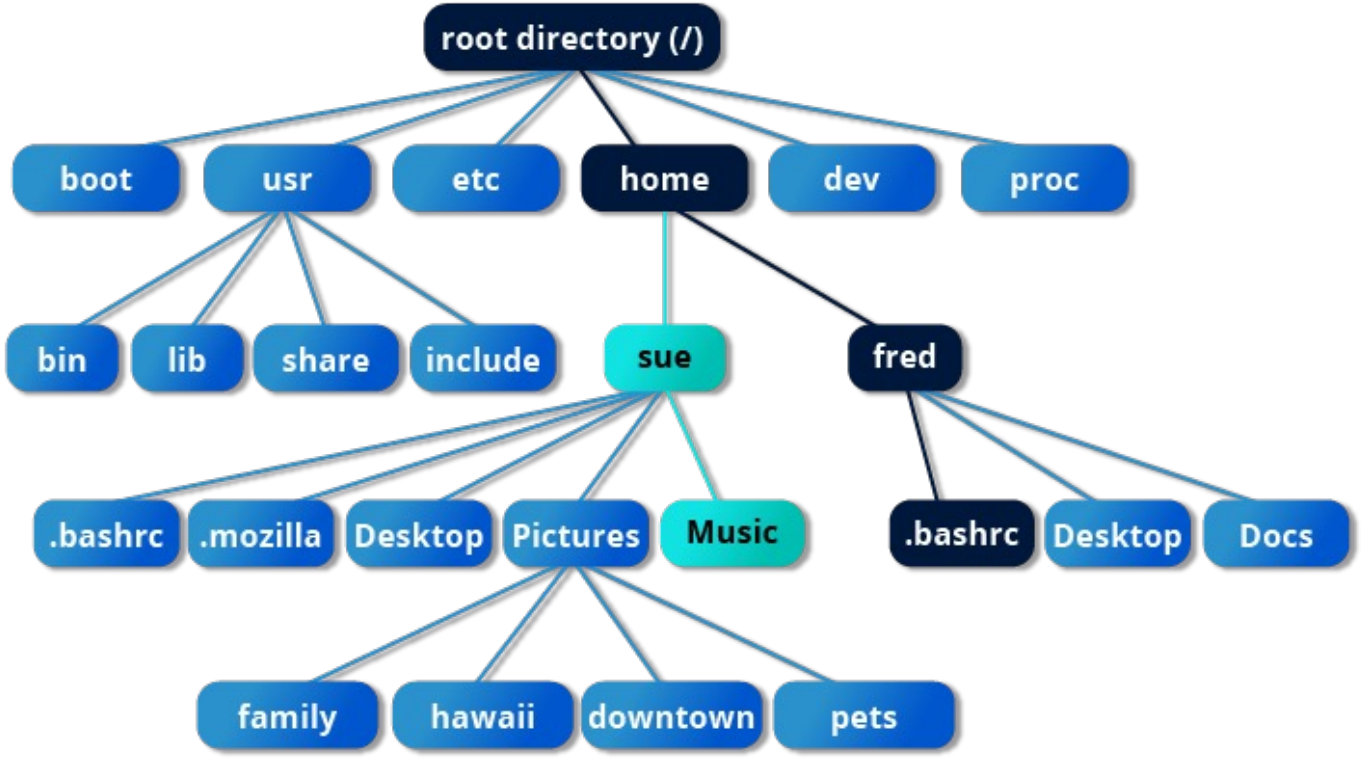
Understanding Paths

- An absolute pathname begins with the root directory (/) and follows the tree, branch by branch, until it reaches the desired directory or file. Absolute paths always start with /.
- A relative pathname starts from the present working directory. Relative paths never start with /.
- Multiple slashes (/) between directories and files are allowed, but all but one slash between elements in the pathname is ignored by the system.
- Most of the time, it is most convenient to use relative paths, which require less typing. Usually, you take advantage of the shortcuts provided by: `.` (present directory), `..` (parent directory), and ~ (your home directory).

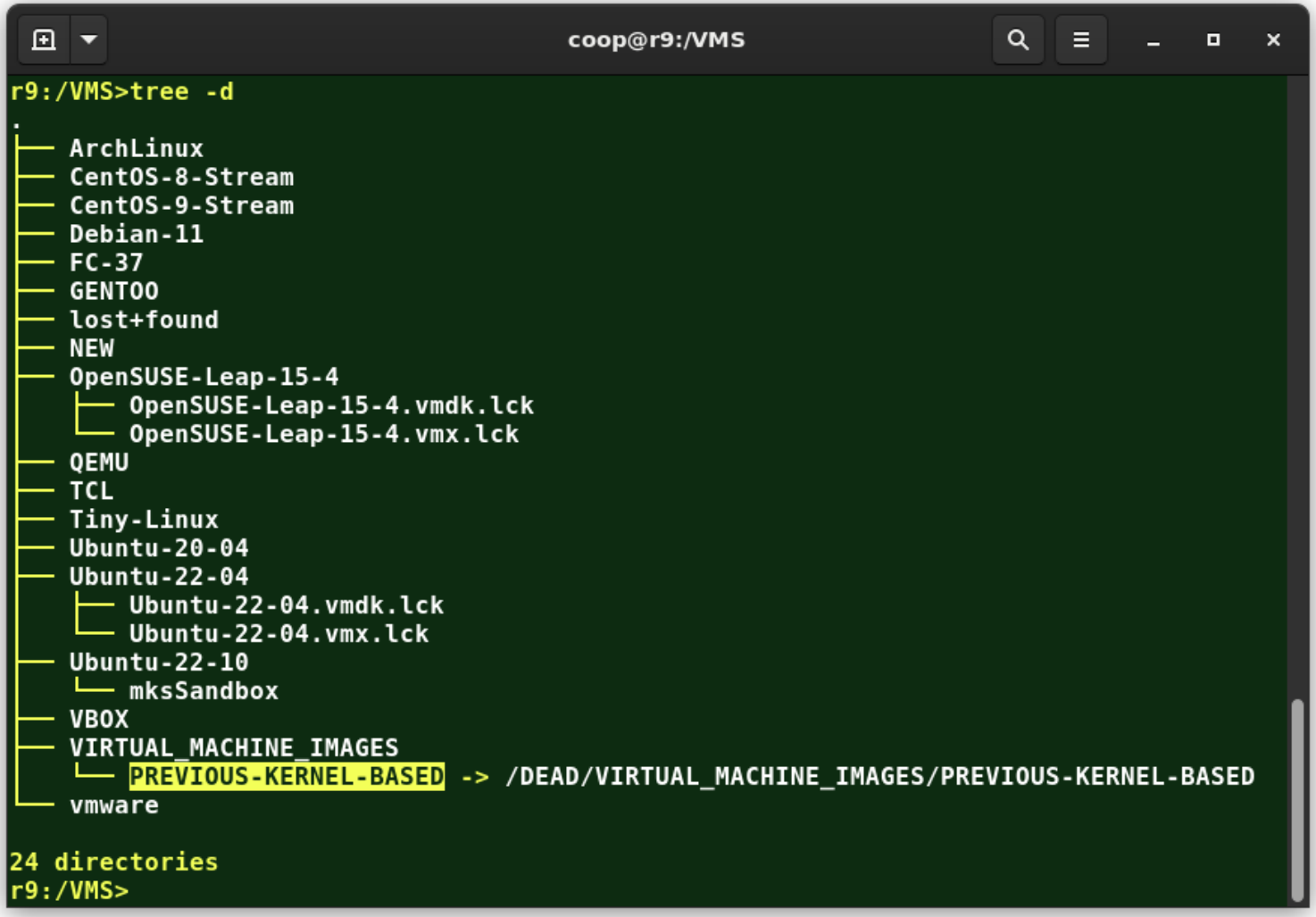
Understanding Paths

Suppose you are currently working in your home directory and wish to move to the `/usr/bin` directory. How can we go there?

- Absolute pathname method = ?
- Relative pathname method = ?



Exploring Filesystem



Single Dash (-) vs Double Dash (--) Options

Features	Single dash	Double dash
Letter or word	-letter `$ ls -l`	-word `$ ls -all`
Chaining	`$ ls -l -a -h` `$ ls -lah`	`--all --human-redable` `--allhumanreadable` is invalid
Passing arguments	No passing argument	`$ grep --context=3 "error" log.txt` `$ grep --context 3 "error log.txt`

Special cases	End of options
	Suppose you try to delete a file named `-file.txt`
	`$ rm -file.txt`
	`$ rm -- -file.txt`
		Standard I/O or previous
		`$ cat -`
		`$ cd -`

Free Linux Playgrounds

- Killercode (1 hour): https://killercoda.com/playgrounds/scenario/ubuntu
- iximiuz Lab (2 hours): https://labs.iximiuz.com/playgrounds?category=linux
- LabEx: https://labex.io/tutorials/linux-online-linux-playground-372915
- GetVM (30 mins): https://getvm.io/playgrounds/vnc-ubuntu-2404
- Linux Containers (30 mins): https://linuxcontainers.org/incus/try-it/

Google Data Center Hackathon

https://docs.google.com/forms/d/e/1FAIpQLSceEHq-OUkeHYAFlJ8bA4NV5dT8lxwS-gXcaGBiMPLZ4j4Scw/viewform

